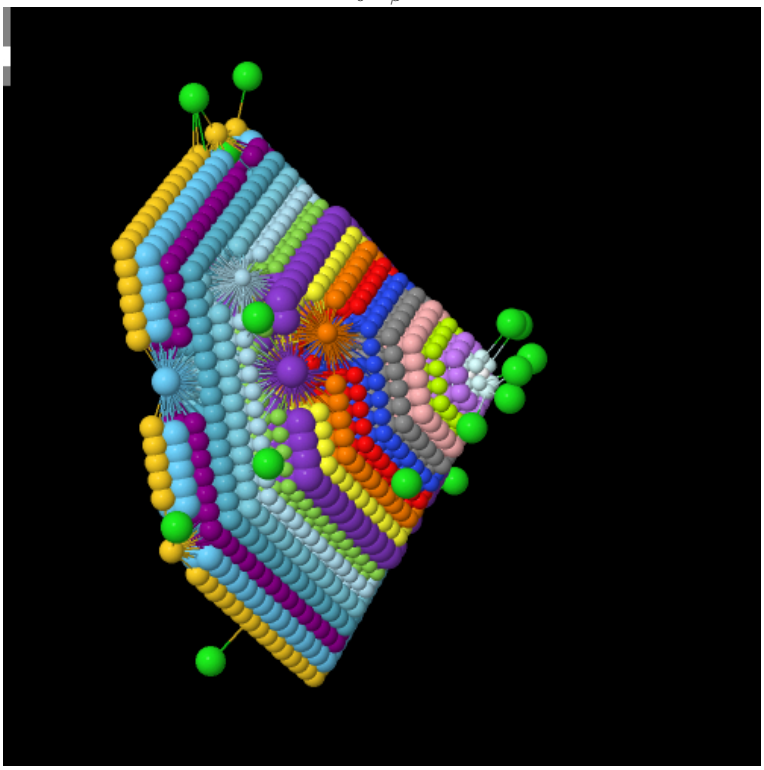
Flawless by Design

The Representation and Construction of Perfects

An Illustrative Explanation to Fermat's Last Theorem and Beal's Conjecture

$${}^{\theta}_{\delta}\Delta^{\Theta}_{\beta}$$



Dempster Welling

Candlemas 2022

Dedication

Thomas Malthus Thomas Hobbes (1588-1679) Pierre Fermat Blaise Pascal

Susan Ann Marie Welling

Sisters Concepta Dorthea Phillips Vera Thomas Regina-Maria [1] and some Others[2]

[1]For the Franciscan nuns from St. Pius X grammar school and elsewhere

[2]Dennis Wick Don Lombard Aden Lombard Yataka Takagi Craig Anderson Michael Canella Richard Stringini

Author's Note

Keeping things as simple as possible, elliptic curves and cyclotomic numbers are avoided. This article does not require advanced mathematics to understand. This article does not make reference to any other work. Representation of perfects and the construction presented show solution. Both conjectures are solved by the same approach. One explanation. The conjectures are put into a form which show that the operation (remove or blend) on two perfects defy exact reconstruction for re-constitution to a perfect. The *proof* is simply from the definition of perfects. The mechanism is derived from a *faithful representation* describing all cases of the constructive model. The mechanism shown demonstrates the impossibility of construction within the constraints. The constraint of having symmetric identical sources. During construction balance must be maintained. When the perfect is constructed one layer at a time, from one source, placing the apex is unconstrained. The mechanism's constraint is a property of bilateral symmetry (bls). For a formation to have bls, the formation is required to have an apex and left/right/mirror symmetric sections flanking center line. The center is a linear arrangement of unit points. The center provides the notion of dimension. The fulcrum is the cornerstone of the constructible object. The apex is the last point placed.

This is the only fact needed for *theorem.* Having two or more identical symmetric sources with which to form the perfect constrains construction to two or more layers at a time which results in the exclusion of the apex. To place apex would require having an imbalance of points (ie $weight \bmod sources \neq 0$) with $\mathrm{L} \approx \mathrm{R}$. The apex and fulcrum change roles when the perfect transforms. In *free* space, the apex and fulcrum are not sensible. In *free* state the the object obtains a new layer of symmetry, a mid plane in the top direction. The center is a grid in even orders and a axial in odd orders.

Abstract

Any perfect ${}^{\theta}\Delta_{\beta}$ can be represented as a bilateral ${}^{\theta}_{\epsilon}\triangleleft\uparrow\triangleright_{\beta-1}$.

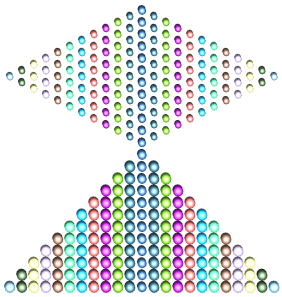
List of Figures

1 a triangular square 12
2 recombination after difference of 3rd order perfects 12
3 recombination after difference of odd order perfects 13
4 flat bls of view of recombination of perfects 13

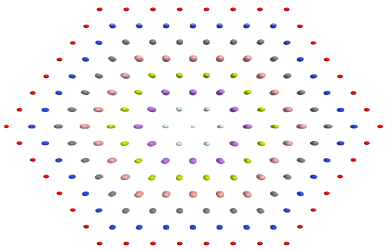
List of Tables

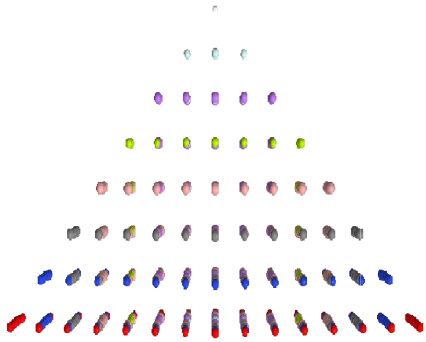
1 Layer Table 14
2 Epsilon Weight Characteristic Table 15
3 Residue Weight Table all tiers : sum of epsilon weights that form core 15

1 Representing Perfects



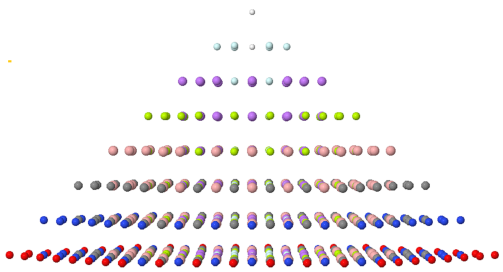
Δ is the perfect symbol. ${}^{\theta}_{\epsilon}\triangleleft\uparrow\triangleright_{\beta-1}$. is a bilateral. θ is the order symbol. β is the dimension symbol. ϵ is the weight characteristic symbol. ${}^{\theta}\Delta_{\beta}$, the perfect is a countable *arrangement* of points and places. Each point or place is assigned a weight. $\beta = \sqrt[\theta]{whole(\Delta)}$ is old school. $\beta - 1$ is the weight of points in the center of odd order Δ. The odd order center consist of 1p unit points. $\beta^2 - 1$ is the weight of points and places in the center of even order Δ. The even order center consist of 1p unit points and 2s1p weight places.



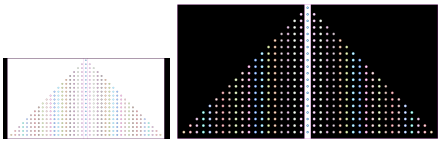


ε epsilon table

	1	2	3	4	5	6	7
2 flat	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7
4	2	8	18	32	50	72	98
5	4	28	92	216	420	724	1148
6	8	92	432	1328	3200	6588	12152
7	16	292	1928	7704	22920	56236	120512
8	32	908	8328	42992	157400	459372	1142288



6s axial 3d model of odd perfects

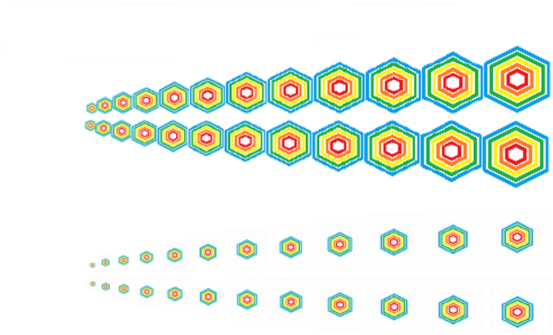


bilateral representations of even and odd perfects.

The apex is the unique point. The center is a stacked arrangement of unit weight points. In even order, the center is a 2d triangular form(area) $(3 + 5 + 7 + \dots) \beta^2 - 1$. In odd order, the center is a 1d linear form(line) $(1 + 1 + 1 + \dots) \beta^1 - 1$. The middle center is the 1d form present in any order. For any order perfect the middle center has weight $\beta - 1$.
The clusters are a hexagonal arrangement of epsilon ϵ weighted points surrounding the middle center in layers with weight that vary based on dimension and order. There are six identical clusters in every perfect $\theta \geq 3$ (symmetry 6s).
Representation as a bls for even and odd reduces explanation to a single case. The middle center has an *even* number of points. *Even* as $\beta - 1 \mod s \equiv 0$. Being *prime* is having one source. 1s. *Symmetry breaking* [3] below order 3 is required to catch all order perfects in the same bls mechanism. In flat 2d, the order two perfect P^2 is a triangle. All points in the triangular formation have unit weight. For even order, the non-middle center points are *places* to put ϵ weighted sub-arrangement. The places have unit weight.
1p - a unit weight point (particle or place).
2s - two-sided ϵ weight.
3s - three-sided ϵ weight.
6s - six-sided ϵ weight.
Sided-ness relates to the number of objects in a symmetric formation.

[3] Symmetry breaking is changing a 3d axial view to a flat 2d view

2 Sub-Polynomials



$$P^1 =_1 C + \alpha$$
$$P^2 =_2 C + \alpha$$
$$P^3 =_3 C +_1 C + \alpha$$
$$P^4 =_4 C +_2 C + \alpha$$
$$P^5 =_5 C +_3 C +_1 C + \alpha$$
$$P^6 =_6 C +_4 C +_2 C + \alpha$$
$$P^7 =_7 C +_5 C +_3 C +_1 C + \alpha$$

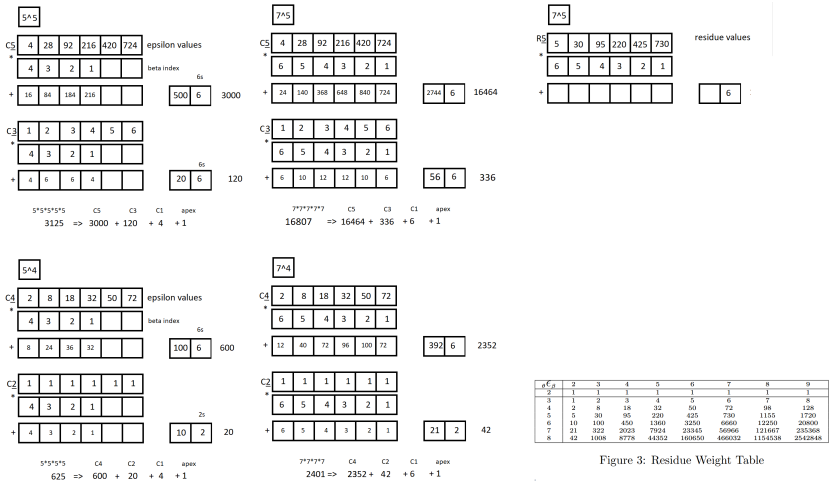
$${}_2C =_2 C_{def} +_1 C \quad \text{only center for even order}$$

$${}_2C_{def} \quad \text{is triangular square with center removed}$$

$$\alpha \quad \text{is the } \textit{weight} \text{ of apex}$$

$${}_\theta C = \sum_\beta {}_\theta S_\epsilon(\beta)$$

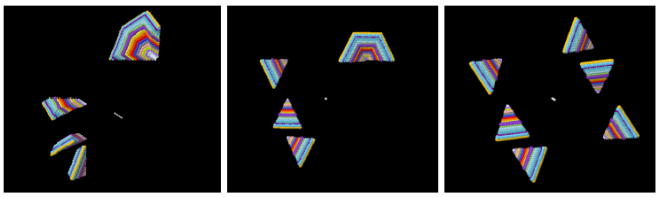
The cluster, petal/wedge 'residue weight vector' is the sum of the tiers of epsilon values for given order.



ϵ_n	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	5	30	95	220	425	730	1155	1720
6	10	100	450	1360	3250	6660	12250	20800
7	21	322	2023	7924	23345	56966	121667	235368
8	42	1008	8778	44352	160850	466032	1154538	2542848

Figure 3: Residue Weight Table

The clusters can be thought of as scaffolding which 'holds' the weight units. The clusters have a wedge form. The structure of the cluster is identical for any size or order. The epsilon ϵ weight characteristic are the only 'conditionals'. The representation is a petal form, a 3 layer stack of flattened wedges; 6s axial to 2s flat with 3 point unit weight.



3 Computational Algorithm

$$H_\Delta = \begin{cases} 0 & \theta \text{ zero,} \\ \Delta(\theta,\beta,\delta) & \theta,\ \beta \text{ and } \delta \text{ whole.} \end{cases}$$

$$\Delta(\theta,\beta,\delta) = (\beta+\delta)\Delta(\theta-1,\beta,\delta) + \delta \sum_{r=1}^{\beta-1} \Delta(\theta-1,r,\delta)$$

$$P^\theta - P^{(\theta-2)} = P^{(\theta-2)}(P^2-1)$$

$$C_\theta = P^\theta - P^{(\theta-2)}$$

$$C_\theta \mod 6 \equiv 0$$

$$C_\theta \quad has \quad six \quad identical \quad parts \quad when \quad \theta{>}2$$

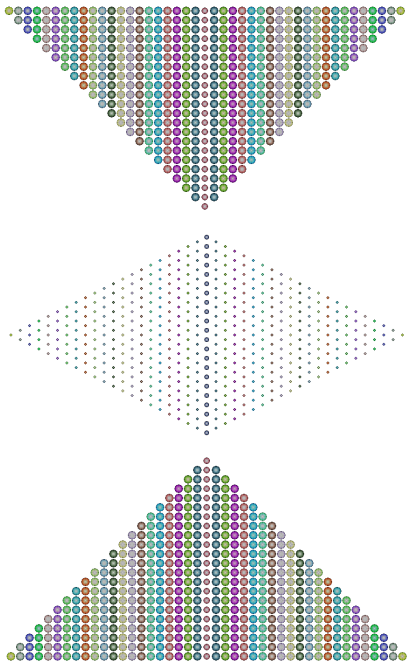
$$S_\theta = \Delta(\theta,\beta,\delta) - \Delta((\theta-2),\beta,\delta)$$

$$\epsilon_\psi = ((\Delta(\theta,\beta+1,\delta) - \Delta(\theta-2,\beta+1,\delta)) - (\Delta(\theta,\beta,\delta) - \Delta(\theta-2,\beta,\delta))$$

$$\epsilon_\theta = \begin{cases} 1 & \theta = 2 \text{ (flat),} \\ \epsilon_\psi/6 & \theta > 2\ . \end{cases}$$

Equations shown here define the $H_p^y e$ mechanism. See Tables 1-3. The mechanism generates the complete set of differences for perfects of all orders. The difference of two perfects is an additive method. The recursive function uses only addition. Perfects are formed from the sub-polynominal clusters. The cluster is shown to have a 6 symmetry. The epsilon weights ϵ_ψ are 6s characteristics of a layer which define the weight of the rings in the cluster.

4 Construction



The Δ^{whole} perfect is constructed as layers of concentric rings and an apex. The construction starts by placing a *fulcrum*. Growth or expansion cycle is *surround* [4] to contain and increase all rings.
The solid perfect $\theta \geq 3$ is modeled as a 3d hexagonal pyramid. The components are arrangements of points. Density increases in rings across the layer. The number of points in the arrangement is the weight of the component. To measure is to count. The tally is weight. To construct is model growth. Growing the bls representation is two-sided. The left AND right sides grow.
The form of representation for all orders is a triangular form. The ring weights(ϵ_{ring}) are arranged in a bi-laterally symmetric representation. Two kinds are possible. Odd θ or even θ.

- $\Delta^{odd} = \sum_r^{odd} C_\beta^r + (\beta^1)_{def} + \alpha$ 6s-axial (linear center)
- $\Delta^{even} = \sum_r^{even} C_\beta^r + (\beta^2)_{def} + \alpha$ 2s-bilateral (triangular square center)

${\beta^1}_{def}$ is sum of points in center. The apex is not in center. ${\beta^2}_{def}$ is sum of points in apex-less triangular square. The apex is not a layer.

[4] To surround in model view is to place a ring around the growing 3d form. To surround in bls view is to add ${}_\theta\epsilon_\beta$ weight far left and far right.

ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6	ϵ_7	ϵ_8
apex								
α_0	ω_1							
α_0	ω_1	ω_2						
α_0	ω_1	ω_2	ω_3					
α_0	ω_1	ω_2	ω_3	ω_4				
α_0	ω_1	ω_2	ω_3	ω_4	ω_5			
α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6		
α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	
α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8

6s axial where ω_x are the x dimension 6p weight units

ϵ_8	ϵ_7	ϵ_6	ϵ_5	ϵ_4	ϵ_3	ϵ_2	ϵ_1	ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6	ϵ_7	ϵ_8
								apex								
							ω_1	α_0	ω_1							
						ω_2	ω_1	α_0	ω_1	ω_2						
					ω_3	ω_2	ω_1	α_0	ω_1	ω_2	ω_3					
				ω_4	ω_3	ω_2	ω_1	α_0	ω_1	ω_2	ω_3	ω_4				
			ω_5	ω_4	ω_3	ω_2	ω_1	α_0	ω_1	ω_2	ω_3	ω_4	ω_5			
		ω_6	ω_5	ω_4	ω_3	ω_2	ω_1	α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6		
	ω_7	ω_6	ω_5	ω_4	ω_3	ω_2	ω_1	α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	
ω_8	ω_7	ω_6	ω_5	ω_4	ω_3	ω_2	ω_1	α_0	ω_1	ω_2	ω_3	ω_4	ω_5	ω_6	ω_7	ω_8

2s bls where ω_x are the x dimension 3p weight units

For even order, the circled positions in the formation are 1 unit weighted places.
This condition here[5] requires one to break symmetry to cover all 1p points and places in the 2nd order tier.

ϵ_6	ϵ_5	ϵ_4	ϵ_3	ϵ_2	ϵ_1	ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6
						3						
					1	0	1					
				2	1	0	1	2				
			3	2	1	0	1	2	3			
		4	3	2	1	0	1	2	3	4		
	5	4	3	2	1	0	1	2	3	4	5	
6	5	4	3	2	1	0	1	2	3	4	5	6

ϵ_6	ϵ_5	ϵ_4	ϵ_3	ϵ_2	ϵ_1	ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6
						4						
					2	0	2					
				8	2	0	2	8				
			18	8	2	0	2	8	18			
		32	18	8	2	0	2	8	18	32		
	50	32	18	8	2	0	2	8	18	32	50	
72	50	32	18	8	2	0	2	8	18	32	50	72

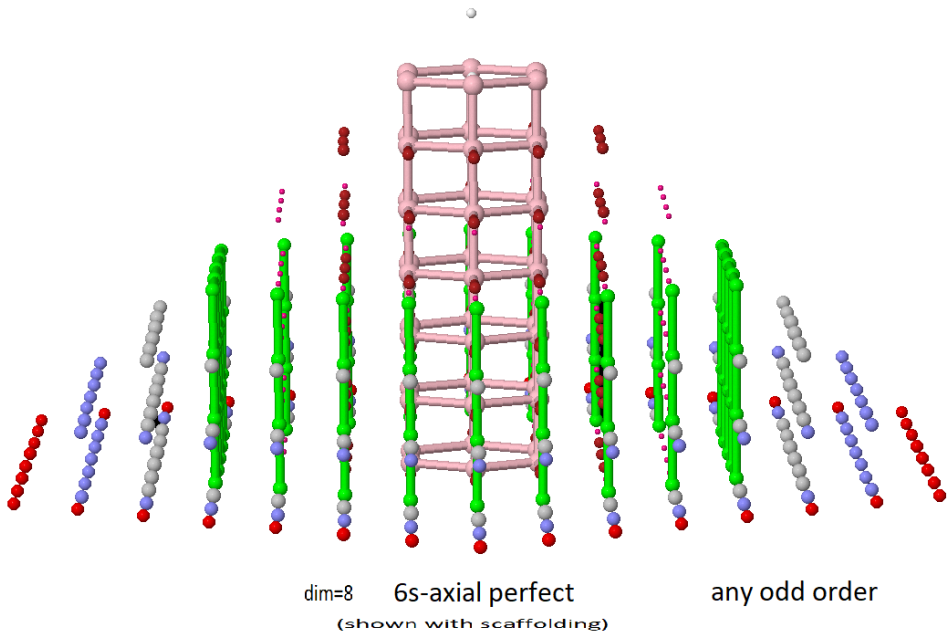
ϵ_6	ϵ_5	ϵ_4	ϵ_3	ϵ_2	ϵ_1	ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6
						5						
					4	0	4					
				28	4	0	4	28				
			92	28	4	0	4	28	92			
		216	92	28	4	0	4	28	92	216		
	420	216	92	28	4	0	4	28	92	216	420	
724	420	216	92	28	4	0	4	28	92	216	420	724

ϵ_6	ϵ_5	ϵ_4	ϵ_3	ϵ_2	ϵ_1	ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6
						6						
					8	0	8					
				92	8	0	8	92				
			432	92	8	0	8	92	432			
		1328	432	92	8	0	8	92	432	1328		
	3200	1328	432	92	8	0	8	92	432	1328	3200	
6588	3200	1328	432	92	8	0	8	92	432	1328	3200	6588

ϵ_6	ϵ_5	ϵ_4	ϵ_3	ϵ_2	ϵ_1	ϵ_0	ϵ_1	ϵ_2	ϵ_3	ϵ_4	ϵ_5	ϵ_6
						7						
					16	0	16					
				292	16	0	16	292				
			1928	292	16	0	16	292	1928			
		7704	1928	292	16	0	16	292	1928	7704		
	22920	7704	1928	292	16	0	16	292	1928	7704	22920	
56236	22920	7704	1928	292	16	0	16	292	1928	7704	22920	56236

[5] something somewhere

5 Explanation



The question, "Can one obtain a perfect from the sum or the difference of two perfects? Yes for $\theta = 2$"

The question, "Can one construct any perfect from two identical sources? Only if sources are unit points. No decimating. "

The question, "Why is GFLT(3;3 , 6;3 , 3;5) 243 - 216 = 27 , 243 - 27 = 216 , 27 + 216 = 243 not a perfect triple? 2 2s3p sources. Missing center. Improper form. Impossible to construct. A 5s1p arrangement can be formed. 5*43+1 "

The question, "Can one devise a mechanism which faithfully represents the *weight* of points in a perfect and disallows the construction from identical sources of 2s3p points? Yes indeed."

A perfect is a countable arrangement of points. A perfect is a object that has form and weight. It is a growth model.

FLT(A,B,C,n) supposedly proves for perfects of the same degree.

GFLT(A;x , B;y , C;z) shows that for any bls form, which all perfects are, construction from multiple identical sources disallows exactness; at best, off by one. The closest one can get to reconstructing one perfect out of two is one extra or one missing point.

Perfects meet the necessary bilateral symmetry conditions[6] to conclude exact reconstruction not possible after difference operation.

5.1 Fermat's conjecture

Fermat's conjecture does not even rate as a brain teaser once one sees what happening in this simple model.

[6] one middle center point per layer, the middle center points weight of one unit, non - center points 3p or 6p weights and the balance of weight above fulcrum.

5.2 Beal's conjecture

The proof tree for Beal's conjecture has few branches. For perfects which are short and squat (3^9) compared to those which are tall and thin (12^4) differences are tricky to reconcile. One discovers that *negatives* arise in the context of the difference operation. The *negatives* are points outside the intersection in the middle center. Reconciled by procedure : a left-side particle pairs with the *negative* and removed : the mirror right-side particle moves to the *negative's* place. This maintains notion of balance. This recombination process requires one to fracture some pairs of 2s3p particles in a petal. After all *negatives* have been dealt with, an attempt is made to reassemble to perfect form. In the vacated places, a new perfect needs to be formed out of the decimated particles from usable rings. (see Fig. 1). Once recombination process is completed, the entire arrangement has to be of perfect form as well.

It is a given that for some values a center-less core is constructible. The mid-center is of the same form in any perfect. One cannot construct a perfect from two different order perfects because the apex cannot be placed above mid-center.

It is stipulated that one can find a triple which has a complete core. The center is constructible. No prohibitions, but still apex-less. To find a triple where the sum or the difference can form a core(apex-less perfect) is less difficult. For instance, Ramujon's flt counter example[7] It is off by one. For gflt, the recombination process is intractable. Do not go down that rabbit hole. Futile. Once the candidate triple's core is found it is still not possible to *complete* formation. The centers *weight* mod *sources* = 0. To model a perfect as a hexagonal pyramid allows one to *see* the center. For order two, the triangular difference and the triangle formed in the vacancy are of even dimension. No decimation of particles is required.
Is decimation being required a strong enough condition to halt search? Perhaps. No. Decimation must be considered, if not, no go. Decimation starts at the largest ring.
perfect A 2S area + x line + apex
perfect B 2T area + y line + apex
Is the fact that addition and subtraction requires like things a strong enough condition to halt search? No.

Adding together would result in 2 apexes.
Subtracting together would result in 0 apexes.
If the difference has no center, no 1p points, can the search be halted?
The condition that the center particles must be unit weight is intrinsic to the mechanism. Rings and layers provide the structure. But nothing else is of consequence except the geometric two-sidedness of the point source resulting from the difference operation on two perfect objects.

[7] i.e., (e.g., $1782^{12} + 1841^{12} = 1922^{12}$)

5.3 PrimaFacia

One can see that neither conjecture can have a counter example even if a common factor exists since the difference is two-sided.

Show. Given two perfects A B can a perfect C be constructed from there summation?

Show. Given two perfects C B can a perfect A be constructed from there difference?

Step 1: Represent A, B and C as flat BLS.

Step 2: Remove centers.

Step 3: Perform operation on objects without centers.

Step 4: Show two-sidedness. Stipulate construction of core possible.

Step 5: Use two layer trick. The modulo trap works on every bls object, by definition of the constraint. 2 w + 1

Step 6: QED shown.

The difference of two perfects generates two identical sources with which to finish the reformation.
Two sources excludes constructing a complete exact perfect.
In context of the mechanism, and the way points are counted, perfects have four parts, two sides left/right which are identical, the center and the unique apex.
The number of geometric sources and weight of x-symmetric units determine the points in the middle center.
Given two sources, the number of points in the middle center is 2x. The l/r sides have 2s and apex has 1 for 2s + 2x + 1. Impossible to place apex, uneven.

6 Illustrations

A triangular square is $(1 + 3 + 5 + \ldots)$ formation. For 2nd order perfects AA plus BB can equal CC. The reason is that all weight units are one. For 3rd or higher order, off center weights are not unit.

Figure 1: a triangular square

A 3d model of a ${}^{3}Perfect$ after a difference operation. Divet is filled in with points from brim. Six identical pieces are being reformed with points of the same piece. Points are confined to the piece where originated or put in center. Decimation is not required here or even possible. Available points are 1p units in the 3d model. A single 1p point is placed in the center for each of the six pieces to maintain symmetry. This forces one to fill in six layers at a time (6 6 6 6 6 6 . . .). The apex is placed (. . . 6 6 6 1) and a 5 mod 6 residue is leftover. If one has a fulcrum to place first (1 6 6 6 6 ... 6 6 6), after fill every six layers a perfect is formed. But the supply of points is $x \mod 6 \equiv 0$ leaving 1p short or 5p over.

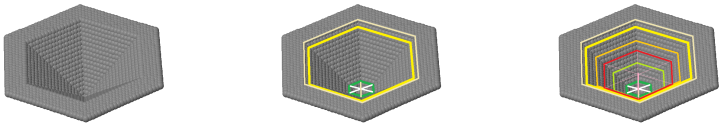


Figure 2: recombination after difference of 3rd order perfects

A 3d model of a $^{Odd}Perfect$ after a difference operation. Divet is filled in with points from brim. Six identical pieces are being reformed with points of the same piece. Points are confined to the piece where originated or put in center. Decimation is required here if $order \geq 4$. Available points are epsilon units in this 3d model. A single 1p point is placed in the center for each of the six pieces to maintain symmetry. This forces one to fill in six layers at a time (6 6 6 6 6 6 . . .). The apex is placed (. . . 6 6 6 1) and a 5 mod 6 residue is leftover. If one has a fulcrum to place first (1 6 6 6 6 ... 6 6 6), after fill every six layers a perfect is formed. But the supply of points is $x \mod 6 \equiv 0$ leaving 1p short or 5p over.

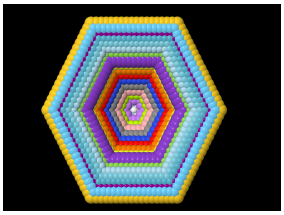


Figure 3: recombination after difference of odd order perfects

A flat 2d representation of a $^{any}Perfect$ after a difference operation. A perfect is formed above with points from the lower edges. Two identical pieces are being reformed with points of the same piece. Points are confined to the side where originated or put in center. Decimation is required here if $order \geq 3$. Available points are 3p epsilon units or 2s1p in this representation. A single 1p point is placed in the center for each of the two pieces to maintain symmetry. This forces one to fill in two layers at a time (2 2 2 2 2 2 2 . . .). The apex is placed (. . . 2 2 2 1) and a 1p point is leftover. If one has a fulcrum to place first (1 2 2 2 2 ... 2 2 2), after fill every two layers a perfect is formed. But the supply of points is $x \mod 2 \equiv 0$ leaving 1p short or 1p over.

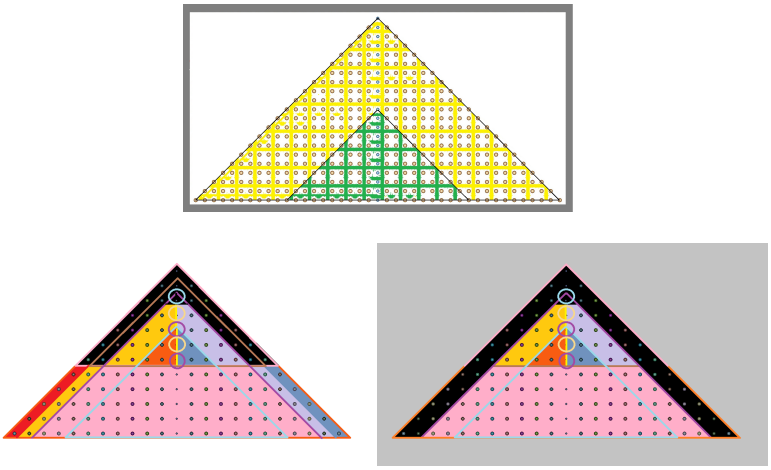


Figure 4: flat bls of view of recombination of perfects

δ 1	0	1	2	3	4	5	6	7	8
1	1	1	1	1	1	1	1	1	1
2	1	3	5	7	9	11	13	15	17
3	1	7	19	37	61	91	127	169	217
4	1	15	65	175	369	671	1105	1695	2465
5	1	31	211	781	2101	4651	9031	15961	26281
6	1	63	665	3367	11529	31031	70993	144495	269297
7	1	127	2059	14197	61741	201811	543607	1273609	2685817
8	1	255	6305	58975	325089	1288991	4085185	11012415	26269505

δ 2	0	1	2	3	4	5	6	7	8
1	2	2	2	2	2	2	2	2	2
2	4	8	12	16	20	24	28	32	36
3	8	26	56	98	152	218	296	386	488
4	16	80	240	544	1040	1776	2800	4160	5904
5	32	242	992	2882	6752	13682	24992	42242	67232
6	64	728	4032	14896	42560	102024	215488	413792	737856
7	128	2186	16256	75938	263552	745418	1817216	3959426	7902848
8	256	6560	65280	384064	1614080	5374176	15097600	37281920	83222784

δ 3	0	1	2	3	4	5	6	7	8
1	3	3	3	3	3	3	3	3	3
2	9	15	21	27	33	39	45	51	57
3	27	63	117	189	279	387	513	657	819
4	81	255	609	1215	2145	3471	5265	7599	10545
5	243	1023	3093	7533	15783	29643	51273	83193	128283
6	729	4095	15561	45927	113553	246519	484785	882351	1509417
7	2187	16383	77997	277749	807159	2019027	4503033	9176457	17390019
8	6561	65535	390369	1673055	5699265	16386591	41367105	94235199	197581665

δ 4	0	1	2	3	4	5	6	7	8
1	4	4	4	4	4	4	4	4	4
2	16	24	32	40	48	56	64	72	80
3	64	124	208	316	448	604	784	988	1216
4	256	624	1280	2320	3840	5936	8704	12240	16640
5	1024	3124	7744	16564	31744	55924	92224	144244	216064
6	4096	15624	46592	116920	258048	515816	953344	1653912	2723840
7	16384	78124	279808	821356	2080768	4704844	9720064	18663628	33734656
8	65536	390624	1679360	5758240	16711680	42656096	98320384	208594080	413204480

δ 5	0	1	2	3	4	5	6	7	8
1	5	5	5	5	5	5	5	5	5
2	25	35	45	55	65	75	85	95	105
3	125	215	335	485	665	875	1115	1385	1685
4	625	1295	2385	4015	6305	9375	13345	18335	24465
5	3125	7775	16775	32525	58025	96875	153275	232025	338525
6	15625	46655	117585	261415	527345	984375	1724905	2868335	4564665
7	78125	279935	823415	2094965	4766585	9921875	19207235	35008265	60651365
8	390625	1679615	5764545	16770655	42981185	99609375	212679265	424216895	798953505

δ 6	0	1	2	3	4	5	6	7	8
1	6	6	6	6	6	6	6	6	6
2	36	48	60	72	84	96	108	120	132
3	216	342	504	702	936	1206	1512	1854	2232
4	1296	2400	4080	6480	9744	14016	19440	26160	34320
5	7776	16806	32736	58806	98976	157926	241056	354486	505056
6	46656	117648	262080	530712	995904	1755936	2939328	4709160	7267392
7	279936	823542	2097024	4780782	9983616	19409046	35551872	61924974	103316352
8	1679616	5764800	16776960	43040160	99934464	213968256	428302080	809965920	1459011840

δ 7	0	1	2	3	4	5	6	7	8
1	7	7	7	7	7	7	7	7	7
2	49	63	77	91	105	119	133	147	161
3	343	511	721	973	1267	1603	1981	2401	2863
4	2401	4095	6545	9919	14385	20111	27265	36015	46529
5	16807	32767	59017	99757	160027	245707	363517	521017	726607
6	117649	262143	531377	999271	1767465	2970359	4780153	7411887	11128481
7	823543	2097151	4782841	9997813	19470787	35753683	62468581	104589961	168762223
8	5764801	16777215	43046465	99993439	214293345	429591071	814051105	1470024255	2546113409

δ 8	0	1	2	3	4	5	6	7	8
1	8	8	8	8	8	8	8	8	8
2	64	80	96	112	128	144	160	176	192
3	512	728	992	1304	1664	2072	2528	3032	3584
4	4096	6560	9984	14560	20480	27936	37120	48224	61440
5	32768	59048	99968	160808	247808	368168	530048	742568	1015808
6	262144	531440	999936	1770832	2981888	4811184	7482880	11272976	16515072
7	2097152	4782968	9999872	19484984	35815424	62670392	105133568	170035832	266338304
8	16777216	43046720	99999744	214352320	429916160	815340096	1474109440	2557125824	4278190080

Table 1: Layer Table

${}_{\theta}\epsilon_1$	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	4	28	92	216	420	724	1148	1712
6	8	92	432	1328	3200	6588	12152	20672
7	16	292	1928	7704	22920	56236	120512	233648
8	32	908	8328	42992	157400	459372	1142288	2522048

Table 2: Epsilon Characteristic Weight Table
single tier

${}_{\theta}\epsilon_\beta$	2	3	4	5	6	7	8	9
2	1	1	1	1	1	1	1	1
3	1	2	3	4	5	6	7	8
4	2	8	18	32	50	72	98	128
5	5	30	95	220	425	730	1155	1720
6	10	100	450	1360	3250	6660	12250	20800
7	21	322	2023	7924	23345	56966	121667	235368
8	42	1008	8778	44352	160650	466032	1154538	2542848

Table 3: Residue Weight Table
all tiers : sum of epsilon weights that form core